

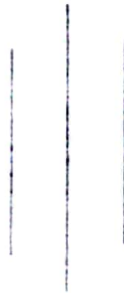
S.N	TABLE OF CONTENT	DATE OF SUBMISSION
1.	Basic Information to Network trouble shooting (part - 1)	2081/03/19
2.	Basic Information to Network trouble shooting (part - 2)	2081/03/29
3.	Research Networking standards	2081/04/02
4.	View Network MAC Addresses	2081/04/10
5.	Observe DNS Resolution	2081/04/18
6.	Building a Basic Router and Switch Network.	2081/04/25
7.	Introduction to Network protocol Analysis	2081/05/08
8.		



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR



Lab Report No:- 1

Title:- ~~Id~~ Basic information to network
troubleshooting (Part -1).

Submitted by:-

Name:- Swaraj Shrestha.

Roll No:- 38

Submitted To:-

Department Of CSIT

Checked by:-

P. Shrestha

Date of submission:- 2081/03/19

TITLE : Basic Information to Network Troubleshooting.

OBJECTIVE :

→ To understand & utilize basic networking commands, including checking network configurations, testing connectivity, understanding IP address, and using network diagnostic tools.

Tools and Software :

- Operating system : ubuntu, windows;
- Command line interface (CLI).

THEORY :

→ In computer networking, understanding and managing network configuration is essential for ensuring proper communication between devices. This lab focuses on utilizing basic networking commands in ubuntu to perform various network diagnostic and configuration tasks. Below is a brief overview of the key concepts and commands covered in this lab.

Network Configuration :-

→ The 'ifconfig' command is used to configure and display network interface parameters. It provides information such as IP address, netmask, and broadcast address. This command is essential for monitoring and managing network interfaces.

• Connectivity x Testing :

→ The 'ping' command checks the connectivity between a local machine and a remote server. By sending ICMP echo request packets to the destination (eg.. 'google.com'). It measures the response time, indicating the status and quality of the network connection.

• IP Address :

→ IP address are unique identifiers assigned to devices on a network. They can be categorized into :

* public (Routable) IP Address : These are accessible over the internet and can be checked using the services like 'whatismyipaddress.com'.

* private (Non-Routable) IP Address : These are used within private networks and are defined by RFC 1918. They are not accessible over the internet and include address ranges like 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.

• Network Address Translation (NAT) :

→ NAT is a technique used to map multiple private IP address to a single public IP address, conserving the number of public IP address required. It allows devices within a private network to communicate with external networks using a shared public IP address.

Tracing Routes :

→ The 'tracroute' command traces the path packet's take from the local machine to a remote destination (eg: google.com). It provides a hop-by-hop account of the journey, which helps in identifying network bottlenecks & failures.

Detailed Network Configuration :

→ The 'ifconfig -a' command displays detailed information about all network interfaces, including the inactive ones. It provides a comprehensive view of the network setup.

Displaying MAC Address :

→ The 'ip link show' command lists all network interfaces along with their MAC (Media access control) addresses. The MAC address is a unique hardware identifier for each network interface.

Viewing the Routing Table :

→ The 'route' or 'ip route show' command displays the system's routing table, which includes the paths data packets take to reach their destination. It shows the default gateway, network interfaces, and routes to different networks.

Commands and Procedures :

1. Checking Network Configuration :

→ Command : 'ifconfig'

- Description : The 'ifconfig' command is used to configure & display network interfaces.

```
surajan@shrestha05surajan:~$ ifconfig
ens20: flags=4096<UP,BROADCAST,MULTICAST> mtu 1500
    ether 7c:84:11:90:11:102 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 11044 bytes 1152184 (1.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11044 bytes 1152184 (1.1 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp0s20f4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.234 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 2405:acc0:1307:f79:6937:32ca:9068:aded prefixlen 64 scopeid 0x0<global>
    inet6 2405:acc0:1307:f79:16 prefixlen 128 scopeid 0x0<global>
    inet6 fe80:16d2:8050:c99d:56ee prefixlen 64 scopeid 0x20<link>
    inet6 2405:acc0:1307:f79:29c4:9097:f30f:ffbe prefixlen 64 scopeid 0x0<global>
    ether 0c:19:a1:3e:a2:9a:f4 txqueuelen 1000 (Ethernet)
    RX packets 342463 bytes 443294919 (443.2 MB)
    RX errors 0 dropped 774 overruns 0 frame 0
    TX packets 88250 bytes 11520160 (11.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. Testing connectivity :

→ Command : 'ping google.com'

- Description : The 'ping' command tests the connectivity between your computer and google's servers.
- Note : use 'ctrl + c' to stop this command.

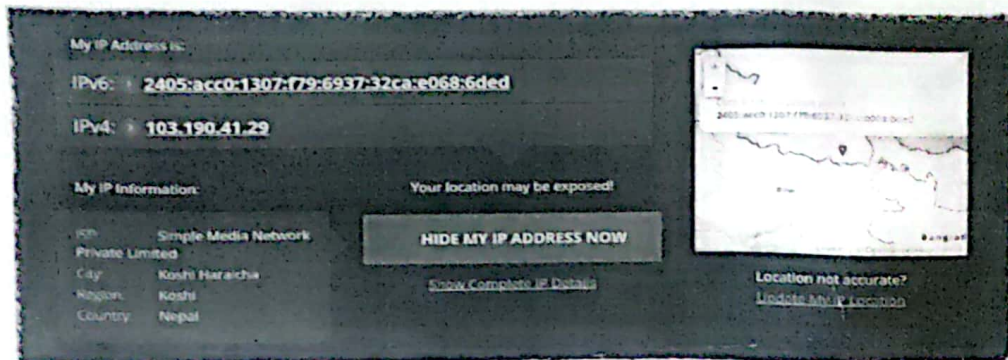
```
surajan@shrestha05surajan:~$ ping google.com
PING google.com (2404:6800:4007:81e::200e) 56 data bytes
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=1 ttl=58 time=53.1 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=2 ttl=58 time=313 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=3 ttl=58 time=50.0 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=4 ttl=58 time=50.3 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=5 ttl=58 time=47.4 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=6 ttl=58 time=52.5 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=7 ttl=58 time=74.7 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=8 ttl=58 time=61.1 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=9 ttl=58 time=53.7 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=10 ttl=58 time=92.6 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=11 ttl=58 time=181 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=12 ttl=58 time=61.7 ms
64 bytes from maa05s23-in-x0e.1e100.net (2404:6800:4007:81e::200e): icmp_seq=13 ttl=58 time=52.3 ms

---
google.com ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12012ms
rtt min/avg/max/mdev = 47.396/84.386/313.497/73.217 ms
```


3. finding Routable Ip Address (public Ip Address):

→ • Description: A public ip address is an ip address that can be accessed over the Internet. These ip address are routable.

Example:



4. Tracing Route:

→ command: 'tracert google.com'

• Description: The 'tracert' command shows the path packets take to reach Google's servers.



5. Detailed Network Configuration :

→ Command : 'ifconfig -a'

- Description : This command displays detailed information about all network interfaces.

- Output Example :

ifconfig -a

6. Displaying MAC Address :

→ Command : 'ip link show'

- Description : The 'ip link show' command displays the MAC addresses of all network interfaces.

```
surajan-shrestha@Surajan-Shrestha:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
   group default qlen 1000 link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp2s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast state DOWN mode DEFAULT
   group default qlen 1000 link/ether 7c:8a:e1:90:1c:02 brd ff:ff:ff:ff:ff:ff
3: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DORMANT
   group default qlen 1000 link/ether 0c:9a:3c:a2:9a:f4 brd ff:ff:ff:ff:ff:ff
```

7. Viewing the Routing Table :

→ Command : 'route -n' or 'ip route show'

- Description : These commands display the system routing table, showing network paths.

```
surajan-shrestha@Surajan-Shrestha:~$ route -n
Kernel IP routing table
Destination    Gateway      Genmask      Flags Metric Ref    Use Iface
0.0.0.0        192.168.18.1 0.0.0.0      UG    600    0      0 wlp0s20f3
192.168.18.0   0.0.0.0      255.255.255.0 U      600    0      0 wlp0s20f3

surajan-shrestha@Surajan-Shrestha:~$ ip route show
default via 192.168.18.1 dev wlp0s20f3 proto dhcp src 192.168.18.234 metric 600
192.168.18.0/24 dev wlp0s20f3 proto kernel scope link src 192.168.18.234 metric 600
```


Discussion :

→ During this lab, we utilized several networking commands to understand and manage network configurations. Each command serves a specific purpose, from displaying network interface configurations to testing connectivity & tracing routes. The 'ifconfig' command provided insight into the network interfaces and their statuses. The 'ping' command helped us verify connectivity with remote servers and 'traceroute' allowed us to trace the path of packets to the destination. By checking the both public & private ip addresses, we understood the difference between routable and non-routable addresses.

Additionally, examining the routing table with the 'route -n' or 'ip route show' commands gave us an overview of the network paths & gateways used for data transmission. Displaying detailed network configurations and MAC addresses further deepened our understanding of the network setup.

CONCLUSION :

→ In this lab session, we explored various networking commands & tools available in OS to manage and diagnose network configurations and connectivity. Understanding these commands is fundamental for troubleshooting and maintaining network systems. By mastering these tools, we can ensure the efficient and reliable network communication, crucial for any networked environment.

Arshia



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR

Lab Report No:- **2 (part - 2)**

Title:- **Basic information to Network troubleshooting**
(part - 2)

Submitted by:-

Name:- **Swajan Shrestha**

Roll No:- **38**

Submitted To:-

Department Of CSIT

Checked by:-

Purnima

Date of submission:- **2081/03/29**

TITLE : Basic Information to Network troubleshooting

OBJECTIVE :

- To understand & use various network troubleshooting Commands.
- To analyze network Configurations & routes.
- To trace the path packets take to reach a destination.
- To utilize online tools for checking AS number and BGP routes.

Software used :

- ubuntu 20.04 LTS
- Terminal

THEORY :

→ Network troubleshooting involves using a variety of commands and tools to diagnose and resolve network issues. These commands provide insights into network configurations, IP address, routing paths and connectivity. Understanding and utilizing these tools is crucial for network administrators to maintain and troubleshoot network performance. Below are some essential commands and their purposes.

Commands used :

1. nslookup :

→ The 'nslookup' command is used to query the Internet domain name servers. It helps in

finding the ip address associated with a domain name.

Example : 'nslookup google.com'

```
surajan-shrestha@Surajan-Shrestha: $ nslookup google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.195.14
Name:   google.com
Address: 2404:6800:4002:826::200e
```

• netstat -r :

→ It displays the IP routing table.

• Example : 'netstat -r'

```
surajan-shrestha@Surajan-Shrestha: $ netstat -r
Kernel IP routing table
Destination      Gateway          Genmask          Flags      MSS Window  irtt Iface
default          _gateway        0.0.0.0          UG         0 0        0 wlp0s20f3
192.168.0.0      0.0.0.0         255.255.240.0    U          0 0        0 wlp0s20f3
```

5. tracert www.ntc.com.np :

→ The 'tracert' command traces the route packets take to reach the destination.

• Example : 'tracert www.ntc.com.np'

```
surajan-shrestha@Surajan-Shrestha: $ tracert www.ntc.com.np
tracert to www.ntc.com.np (202.70.64.2), 30 hops max, 60 byte packets
 1  _gateway (192.168.0.1)  3.377 ms  3.326 ms  3.307 ms
 2  172.25.25.1 (172.25.25.1)  5.549 ms  5.522 ms  5.496 ms
 3  gw-vlan224.nren.net.np (202.52.1.33)  5.492 ms  5.468 ms  5.451 ms
 4  npix-pts.ntc.net.np (198.32.231.15)  6.713 ms  6.697 ms  6.681 ms
 5  snd.core-ptn.core.ntc.net.np (202.70.93.83)  6.715 ms  6.695 ms  6.679 ms
 6  ptn.acc-ptn.core.ntc.net.np (202.70.93.95)  6.571 ms  10.290 ms  10.261 ms
 7  ptn.acc-ptn.ne.acc.ntc.net.np (202.70.93.100)  4.317 ms  4.620 ms  4.581 ms
```


• dig :

→ The 'dig' (Domain Information Groper) command queries DNS name servers for information about host address, mail exchanges, nameservers, and related information.

• Example : 'dig google.com'.

```
surajan-shrestha@Surajan-Shrestha: $ dig google.com
;<<>> DiG 9.18.24-0ubuntu5-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: NOERROR, id: 47261
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 4, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A
;; ANSWER SECTION:
google.com.                 37      IN      A      142.250.195.14
;; AUTHORITY SECTION:
google.com.                 60222   IN      NS      ns2.google.com.
google.com.                 60222   IN      NS      ns3.google.com.
google.com.                 60222   IN      NS      ns4.google.com.
google.com.                 60222   IN      NS      ns1.google.com.
;; ADDITIONAL SECTION:
ns2.google.com.             60789   IN      A      216.239.34.10
ns3.google.com.             19525   IN      A      216.239.36.10
ns4.google.com.             19525   IN      A      216.239.38.10
ns1.google.com.             60721   IN      A      216.239.32.10
;; Query time: 4 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Wed Jul 03 10:36:43 +0545 2024
;; MSG SIZE rcvd: 191
```

• Hostname :

→ The 'hostname' command displays or sets the system's hostname.

• Example : 'hostname'

```
surajan-shrestha@Surajan-Shrestha: $ hostname
Surajan-Shrestha
```


.. mtr :

- The 'mtr' (my Traceroute) command combines the functionality of 'ping' and 'traceroute' and provides a continuous updating view of the network path.
- Example : 'mtr google.com'

My traceroute [v0.95]
Surajan-Shrestha (192.168.1.221) -> google.com (12024-07-03T10:39:27+0545)
Keys: Help Display mode Restart statistics Order of fields quit

Host	Packets			Pings				
	Loss%	Snt	Last	Avg	Best	Wrst	StDev	
1. _gateway	0.0%	43	1.3	6.5	1.2	95.5	17.2	
2. 172.25.25.1	0.0%	43	1.8	2.9	1.4	13.4	3.0	
3. 1.220.166.202.ether.static	4.7%	43	3.0	4.9	1.9	50.3	7.6	
4. be-7-136.40.gwc-ndc-core-0	2.3%	43	3.5	6.1	2.6	74.0	11.7	
5. ae-20-136.41.gwj-htda-core	2.3%	43	3.2	10.6	2.9	158.7	24.7	
6. ae-21-139.41.gwj-btwl-core	0.0%	43	26.3	13.1	4.9	51.5	9.4	
7. ae52-ipt-bhwa-01.wlink.com	2.3%	43	5.3	8.0	5.0	27.1	4.4	
8. (waiting for reply)								
9. 142.250.174.2	0.0%	43	27.5	27.1	25.6	31.1	1.4	
10. 142.250.208.225	2.4%	42	27.0	32.1	26.5	134.5	17.9	
11. 142.251.76.199	2.4%	42	16.5	25.3	16.5	109.8	18.0	
12. del11s21-in-f14.1e100.net	2.4%	42	16.6	18.4	16.3	36.6	3.9	

7. nta.gov.np :

→ The official website of the Nepal Telecommunication Authority, which provides information on ISPs and regulations.

- website : nta.gov.np

8. ICMP protocol :

a. ICMP Echo Request :

- A type of network packet used to test the reachability of a host on an IP network.
- Commonly known as 'ping'.

b. ICMP Echo Reply :

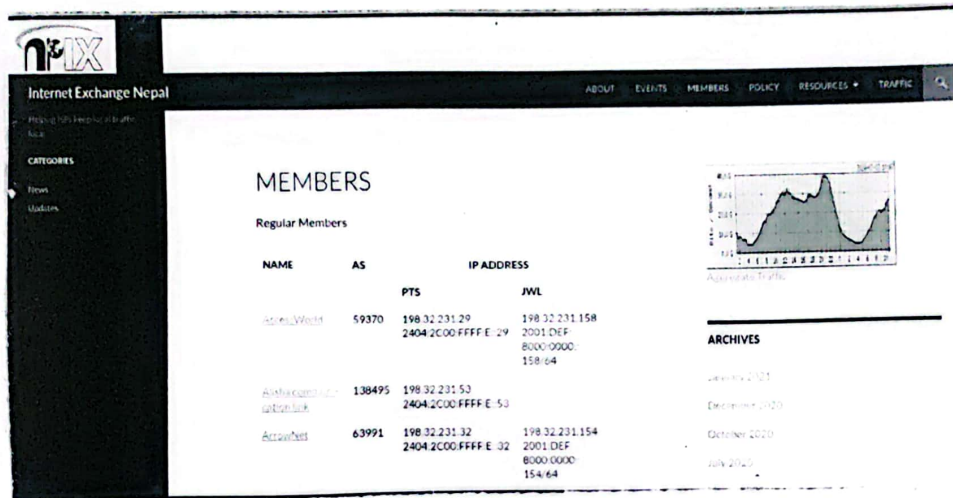
- The response packet sent back to the source after receiving an ICMP Echo request, confirming the

host is reachable.

9. npix.net.np:

→ Nepal Internet Exchange (NPiX) is a local Internet exchange point where various IPs exchange the Internet traffic.

• website : npix.net.np.



10. bgp.he.net:

→ bgp.he.net: Hurricane Electric's BGP Toolkit provides comprehensive information on BGP routing and AS (Autonomous System) numbers. It is a valuable resource for network administrators to lookup BGP routes, check the status of AS numbers, and analyze Internet traffic patterns.

• Example : BGP toolkit - AS23752 or AS59370, etc.

↓ BGP Toolkit - HOME :



HURRICANE ELECTRIC
INTERNET SERVICES

BGP Toolkit Home

Quick Links
BGP Toolkit Home
BGP Prefix Report
BGP Peer Report
Super Traceroute
Super Looking Glass
Exchange Report
Bogon Routes
World Report
Multi Origin Routes
DNS Report
Top Host Report
Internet Statistics
Looking Glass
Network Tools App
Free IPv6 Tunnel
IPv6 Certification
IPv6 Progress
Going Native
Credits
Contact Us

Home

Welcome to the Hurricane Electric BGP Toolkit

You are visiting from 202.166.220.132

Announced as 202.166.220.0/24 (Enterprise Prod)
Announced as 202.166.220.0/20 (WorldLink Communications)
Announced as 202.166.192.0/19 (WorldLink Communications)

Your ISP is AS17501 (WorldLink Communications Pvt Ltd)

Search

↓ AS59370 Access World Tech Pvt. Ltd. :



HURRICANE ELECTRIC
INTERNET SERVICES

AS59370 Access World Tech Pvt. Ltd.

Quick Links | AS Info | Graph v4 | Prefixes v4 | Prefixes v6 | Peers v4 | Peers v6 | Whois | IRR | IX | Traceroute

BGP Toolkit Home
BGP Prefix Report
BGP Peer Report
Super Traceroute
Super Looking Glass
Exchange Report
Bogon Routes
World Report
Multi Origin Routes
DNS Report
Top Host Report
Internet Statistics
Looking Glass
Network Tools App
Free IPv6 Tunnel
IPv6 Certification
IPv6 Progress
Going Native
Credits
Contact Us

Company Website

<http://www.accessworld.net>

Country of Origin

Nepal

Internet Exchanges: 1

Prefixes Originated (all): 4
Prefixes Originated (v4): 3
Prefixes Originated (v6): 1

Prefixes Announced (all): 4
Prefixes Announced (v4): 3
Prefixes Announced (v6): 1

RPKI Originated Valid (all): 4
RPKI Originated Valid (v4): 3
RPKI Originated Valid (v6): 1

RPKI Originated Invalid (all): 0
RPKI Originated Invalid (v4): 0
RPKI Originated Invalid (v6): 0

BGP Peers Observed (all): 3
BGP Peers Observed (v4): 3
BGP Peers Observed (v6): 1



↳ Looking Glass :

→ Looking Glass servers offer network operators and engineers a valuable tool for remotely accessing and troubleshooting their BGP (Border Gateway protocol) configurations. By providing a web-based interface, users can easily execute commands such as ping, traceroute and bgp to diagnose network issues, verify routes and

Discussion:

The lab exercise involved using several network troubleshooting commands to understand & analyze network configurations, routes and connectivity. Using 'nslookup', we resolved domain names to IP addresses, essential for troubleshooting DNS-related issues. The 'ifconfig' command provided details about network interfaces, their IP addresses and other configurations. The 'netstat -r' command displayed the routing table, showing how packets are routed. The 'traceroute' command traced the path packets take to reach a destination.

We explored the NPIX website to understand the role of local internet exchange points in optimizing local traffic flow. The BGP Toolkit on Hurricane Electric's website provided insights into BGP routes and AS numbers. Finally, we explored the looking glass tool, which allows for executing BGP-related commands via a web interface.

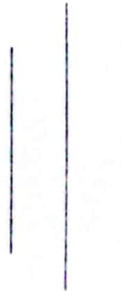
CONCLUSION:
Network troubleshooting commands are essential for diagnosing and resolving network issues, using tools like 'nslookup', 'ifconfig', 'traceroute', etc. which helps identifying and fixing the problems more effectively. Local Internet Exchange like NPIX to optimize traffic, improving performance. Hurricane Electric's BGP Toolkit and the looking glass interface offer valuable insights into global routing & BGP configurations. Mastering these tools is crucial for maintaining an efficient network infrastructure.



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR



Lab Report No:- 2

Title:- Research Networking standards.

Submitted by:-

Name:- suhajan shrestha

Roll No:- 38

Submitted To:-

Department Of CSIT

Checked by:

Purna

Date of submission:- 2081/04/02

TITLE: Research Networking Standards.

OBJECTIVES:

- Research Networking Standards Organizations.
- Reflect on Internet and computer Networking Experiences.

Background / Scenario:

→ using web search engines like Google, research the non-profit organizations that are responsible for establishing international standards for the Internet and the development of internet technologies.

Required Resources:

- Device with Internet access.

STEP 1: Research Networking Standards Organizations.

1. who is Jonathon B. postel and what is he known for?

→ Jonathon B. postel was an American computer scientist known for his pivotal role in the development and administration of the Internet. He was instrumental in the creation of Internet Assigned Numbers Authority (IANA) and managed the allocation of IP address and the assignment of protocol parameters.

2. which 2 related organizations are responsible for managing the top-level domain name space and the root DNS name servers on the Internet?

1. The two related organizations are the Internet Corporation for Assigned Names and Numbers (ICANN) and the Internet Assigned Numbers Authority (IANA).

2. Vinton Cerf has been called one of the main fathers of the Internet. What Internet organizations did he chair or help found? What Internet technologies did he help to develop?

→ Vinton Cerf co-invented the TCP/IP protocols and played a key role in the development of Internet. He was a founding member of the Internet Society (ISOC) and served as its chairman. He also served as the chairman of the Internet Corporation for Assigned Names and Numbers (ICANN).

3. What organization is responsible for publishing Request for Comments (RFC)?

→ The Internet Engineering Task Force (IETF) is responsible for publishing Request for Comments (RFC).

4. What do RFC 349 and RFC 1700 have in common?

→ Both RFC 349 and RFC 1700 deal with the assignment of protocol parameters and are part of the Internet Assigned Numbers Authority (IANA) functions.

5. What RFC number is the ARPANOCKY? What is it?

→ The ARPANOCKY is RFC 527, a humorous RFC published as an APRIL fool's Day joke.

6. Who founded the World Wide Web Consortium (W3C)?

→ Tim Berners-Lee founded the World Wide Web Consortium (W3C).

3, Name 10 WWW standards that the W3C develops and maintains?

→ They are:

↳ HTML, CSS, XML, XHTML, SVG, DOM, MathML, XSLT, SOAP and RDF.

3, where is the Institute of Electrical and the Electronics Engineers (IEEE) headquarters located and what is the significance of its logo?

→ The IEEE headquarters is located in Piscataway, New Jersey, USA. The logo features a diamond-shaped figure which represents the electron tube, symbolizing the early days of electronics and electrical engineering.

10, what is the IEEE standard for the wi-fi Protected Access 2 (WPA2) security protocol?

→ The IEEE standard for the wi-fi Protected Access 2 (WPA2) security protocol is IEEE 802.11i.

1, Is the wi-fi Alliance a non-profit standards organization? What is their goal?

→ Yes, the wi-fi Alliance is a non-profit standards organization. Their goal is to drive the adoption and evolution of wi-fi globally, ensuring interoperability, security and reliability.

12, who is Hamadou Toure?

→ Hamadou Toure is a malian telecommunications engineer who served as the Secretary-General of the International Telecommunication Union (ITU) from 2007 to 2014.

3, what is the International Telecommunication union (ITU) and where is it headquartered?

→ The International Telecommunication union (ITU) is a specialized agency of the United Nations responsible for issues related to information and communication technologies. It is headquartered in Geneva, Switzerland.

4, Name the three ITU sectors.

→ a, ITU - T (Telecommunication Standardization)

b, ITU - R (Radio Communication)

c, ITU - D (Telecommunication Development)

5, what does the RS in RS-232 stands for and which organization introduced it?

→ The RS in RS-232 stands for "Recommended Standard". It was introduced by the Electronic Industries Alliance (EIA).

16, what is Spacewire?

→ Spacewire is a high-speed communications protocol for use onboard spacecraft, facilitating the transfer of data between sensors, processors and storage devices.

17, what is the mission of the ISOC and where are its headquarters located?

→ The mission of the Internet Society (ISOC) is to ensure the open development, evolution & use of the Internet for the benefit of all people throughout the world. Its headquarters are located in the Reston, Virginia, USA and Geneva, ~~Switzerland~~ Switzerland.

18, what organizations does the IAB oversee?
→ The Internet Architecture Board (IAB) oversees the Internet Engineering Task force (IETF) and the Internet Research Task force (IRTF).

19, what organization oversees the IAB?
→ The Internet Society (ISOC) oversees the Internet Architecture Board (IAB).

20, when was the ISO founded and where are its headquarters located?

→ The International Organization for Standardization (ISO) was founded on 23 February 1947. Its headquarters are located in Geneva, Switzerland.

STEP - 2 : Reflect on Internet and Computer Networking experiences.

1, How do the internet standards allow for the greater commerce? what potential problems could we have if we did not have the IEEE?

→ Internet standards allows for greater commerce by ensuring interoperability and compatibility across different systems and networks, facilitating global communication and trade. Without the IEEE, we would face significant challenges in developing unified and reliable standards for electronics, communications and computing leading to incompatibility and inefficiency in technology development.

What potential problems could we have if we did not have the W3C?

→ Without the W3C, the development and maintenance of web standards would lack coordination, leading to fragmentation of web technologies. This would result in compatibility issues across different web browsers and platforms, hindering the seamless user experience and innovation on the world wide web.

What can we learn from the example of the Wi-Fi Alliance with regard to the necessity of networking standards?

→ The example of the Wi-Fi Alliance highlights the importance of industry collaboration in developing and promoting networking standards. Such standards ensure interoperability, enhance security, and promote widespread adoption of technologies, ultimately benefiting consumers and driving the innovation.

Conclusion:

Researching networking standards organizations and their contributions provides valuable insights into the infrastructure and governance of the Internet. These organizations play a crucial role in ensuring the interoperability, security, and the continuous evolution of Internet technologies, which are fundamental to modern communication and commerce.



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR

Lab Report No:- 3

Title:- view Network Device MAC Addresses.

Submitted by:-

Name:- swarajan shrestha.

Roll No:- 38

Submitted To:-

Department Of CSIT

Checked by:-

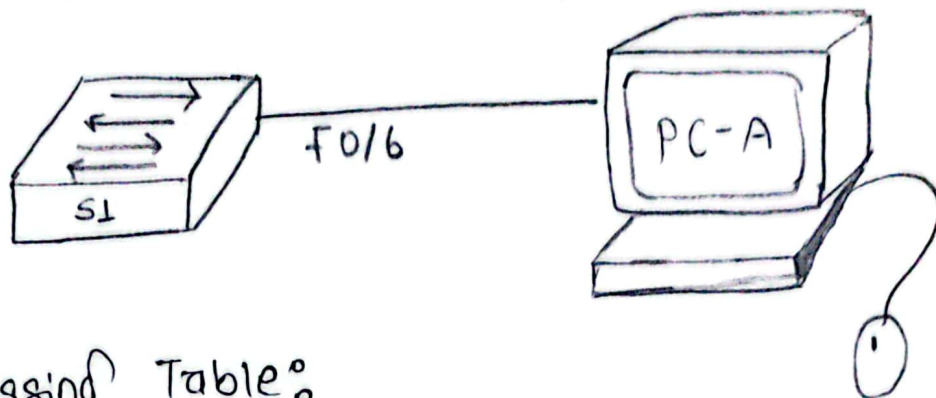
P. K. Shrestha

Date of submission:- 2081/04/10

TITLE: view Network Device MAC Addresses

OBJECTIVES:

1. Configure devices and verify network connectivity.
2. Display, describe and analyze Ethernet MAC address on network devices.



Addressing Table:

Device	Interface	Ipaddress	Subnet mask	Gateway
S1	VLAN 1	192.168.1.2	255.255.255.0	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Required Resources:

- a, 1 switch (Cisco 2960 with Cisco IOS Release 15.2 (2) lanbasek9 image or comparable)
- b, 1 PC (Windows with a terminal emulation program, such as Tera Term)
- c, Console cable to configure the Cisco switch via console.
- d, Ethernet cables as shown in the topology.

THEORY:

→ Every device on an Ethernet LAN is identified by a Layer 2 MAC (Media Access Control) address. This address is a unique identifier assigned to network interfaces for communications on the physical network segment. The MAC address is

embedded into the network interface card (NIC) by the manufacturer and is stored in its firmware, making it a critical component in network communications.

MAC addresses are 48-bits long and are expressed as 12 hexadecimal digits. These digits are often displayed in six sets of two, separated by dashes, colons, or periods (e.g. 00:1A:2B:3C:4D:5E). The structure of a MAC address consists of two parts:

1. Organizationally Unique Identifier (OUI): The first 24-bits (or first three octets) which identify the manufacturer of the NIC.
2. Network Interface Controller (NIC): The last 24 bits (or last three octets) which uniquely identify the NIC.

→ understanding MAC addresses is essential for several reasons:

- Device Identification: Each device on a network has a unique MAC address, making it possible to identify and differentiate between devices.
- Network Management: Network administrators use MAC addresses help in tracing the path of data packets and identifying points of failure or unauthorized devices.
- Security: MAC addresses are used in security protocols to validate device authenticity and prevent unauthorized access.

In this lab, we will explore and analyze the components that make up a MAC address, and how to retrieve this information on a switch and a PC. By configuring devices, verifying connectivity, and analyzing MAC address, we gain a deeper understanding of network communications and device management.

OBSERVATIONS :

↳ Instructions :

part 1 : Configure Devices & verify connectivity.

1, cable the network as shown in topology.

- Attach the devices shown in the topology & cable as necessary.
- power on all the devices in the topology.

2, Configure the IPv4 address for the PC.

- Configure the IPv4 address, subnet mask, and default gateway address of PC-A.
- from the command prompt on PC-A, ping the switch address.

command used :

* Ping 192.168.1.2

Q, were the pings successful? explain.

→ Yes, the pings should be successful if the network is properly configured. A successful ping indicates that PC-A can communicate with the switch, confirming that both devices are correctly configured with their IP addresses and connected via the network.

3, Configure basic settings for the switch.

- console into the switch and enter global configuration mode.

↳ switch > enable

switch #

switch (config) #

- Assign a hostname to the switch based on addressing table.

↳ switch (config) # hostname S1

- Disable DNS lookup.

↳ S1 (config)

- Configure and enable the SVI interface for VLAN 1.

↳ S1 (config) # int vlan 1

S1 (config-if) # ip address 192.168.1.2

S1 (config-if) # no shutdown

S1 (config-if) # end

4, verify network connectivity.

- ping the switch from PC-A.

↳ ping 192.168.1.2

Q, were the pings successful?

→ yes, the pings were successful.

Part 2: Display, Describe and analyze Ethernet MAC Addresses.

5, Analyze the MAC address for the PC-A NIC.

- use the 'ipconfig /all' command to view the MAC address of your NIC.

↳ ipconfig /all

a. Using the output from the `show ipconfig /all` command, answer the following questions

1. `show ipconfig /all`

Output omitted

2. Output omitted

3. Output omitted

4. Output omitted

5. Output omitted

6. Output omitted

7. Output omitted

8. Output omitted

9. Output omitted

10. Output omitted

11. Output omitted

12. Output omitted

13. Output omitted

14. Output omitted

1. IP Address: 192.168.1.100

2. Subnet Mask: 255.255.255.0

3. Default Gateway: 192.168.1.1

4. DNS Servers: 192.168.1.1

5. DHCP Enabled: No

6. IP Address: 192.168.1.100

7. Subnet Mask: 255.255.255.0

8. Default Gateway: 192.168.1.1

9. DNS Servers: 192.168.1.1

10. DHCP Enabled: No

Q, what is the OUI portion of the MAC address for this device?

The OUI (Organizationally Unique Identifier) portion of the MAC address is the first three bytes (six hexadecimal digits). For example, in the MAC address '5C-26-0A-24-2A-60', the OUI portion is '5C-26-0A'.

Q, what is the serial number portion of the MAC address for this device?

The serial number portion of the MAC address is the last three bytes (six hexadecimal digits). In the MAC address '5C-26-0A-24-2A-60', the serial number portion is '24-2A-60'.

Q, using the OUI lookup tool, find the name of the vendor that manufactured this NIC.

To find the vendor name using the OUI '5C-26-0A', you can use the OUI lookup tool at IEEE OUI lookup. According to the OUI lookup, '5C-26-0A' is registered to Intel Corporation.

1, Identify the OUI and serial number portion of the MAC address for the NIC of PC-A.
→ The OUI portion of the MAC address for the NIC of PC-A, if using example '5c-26-0a-24-2a-60', would be '5c-26-0a'.

→ The serial number portion of the MAC address for the NIC of PC-A would be '24-2a-60'.

2, Identify the name of the vendor that manufactured the NIC of PC-A.

→ Using the OUI '5c-26-0a' from the MAC address of PC-A, the vendor is Intel Corporation.

Analyze the MAC address for the s1 Fa/6 interface.
Use the 'show interface vlan 1' command to find the MAC address information.

```
S1# show interfaces vlan 1
```

```
Vlan1 is up, line protocol is up
```

```
Hardware is EtherSVI, address is 001b.1c6d.8f40 (bia 001b.1c6d.8f40)
```

```
Internet address is 192.168.1.1/24
```

```
MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
```

```
reliability 255/255, txload 1/255, rxload 1/255
```

```
Encapsulation ARPA, loopback not set
```

```
Keepalive not supported
```

```
ARP type: ARPA, ARP Timeout 04:00:00
```

```
Last input never, output 00:14:51, output hang never
```

```
Last clearing of "show interface" counters never
```

```
Input queue: 0/75/0/0 (size/max/drops/flushes): Total output drops: 0
```

```
Queueing strategy: fifo
```

```
Output queue: 0/40 (size/max)
```

```
5 minute input rate 0 bits/sec, 0 packets/sec
```

```
5 minute output rate 0 bits/sec, 0 packets/sec
```

```
0 packets input, 0 bytes, 0 no buffer
```

```
Received 0 broadcasts (0 IP multicasts)
```

```
0 runs, 0 giants, 0 throttles
```

```
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
```

```
34 packets output, 11116 bytes, 0 underruns
```

```
0 output errors, 2 interface resets
```

```
0 unknown protocol drops
```

```
0 output buffer failures, 0 output buffers swapped out
```


> what is the MAC address for VLAN 1 on S1?

• The MAC address for VLAN 1 on S1 is 001b.0c6d.8f40

> what is the MAC serial number for VLAN 1?

• The MAC serial number for VLAN is 0c6d.8f40.

• Is derived from the full MAC address, where the last six hex digits represent the serial number portion.

> what does BIA stand for?

• BIA stands for Burned-In Address. This is the permanent MAC address that is assigned to a network device at the factory.

> why does the output show the same MAC address twice?

• The output shows the same MAC address twice because the BIA is the default MAC address for the interface. When you view the interface details

shows both the BIA and the current MAC address, which are the same by default unless the MAC address has been manually changed.

> what layer 2 addresses are displayed on S1?

• The layer 2 addresses displayed on S1 are:

- 001b.0c6d.8f40 (for VLAN 1)

- 5c26.0a24.2a60 (for another device on VLAN 1)

Q, what layer 3 addresses are displayed on s1?

→ The layer 3 addresses displayed on s1 are:

- 192.168.1.2 (associated with the MAC address 001b.0c6d.8f40)
- 192.168.1.3 (associated with the MAC address 5c26.0a24.2a60)

7, view the MAC addresses on the switch.

→ use the 'show mac address-table' command on s1.

```
S1# show mac address-table
```

Mac Address Table

Vlan	Mac Address	Type	Ports
---	-----	-----	-----
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0180.c200.0000	STATIC	CPU
All	0180.c200.0001	STATIC	CPU
All	0180.c200.0002	STATIC	CPU
All	0180.c200.0003	STATIC	CPU
All	0180.c200.0004	STATIC	CPU
All	0180.c200.0005	STATIC	CPU
All	0180.c200.0006	STATIC	CPU
All	0180.c200.0007	STATIC	CPU
All	0180.c200.0008	STATIC	CPU
All	0180.c200.0009	STATIC	CPU
All	0180.c200.000a	STATIC	CPU
All	0180.c200.000b	STATIC	CPU
All	0180.c200.000c	STATIC	CPU
All	0180.c200.000d	STATIC	CPU
All	0180.c200.000e	STATIC	CPU
All	0180.c200.000f	STATIC	CPU
All	0180.c200.0010	STATIC	CPU
All	ffff.ffff.ffff	STATIC	CPU
1	5c26.0a24.2a60	DYNAMIC	Fa0/6

Total Mac Addresses for this criterion: 21

Q, Did the switch display the MAC address of pc-A? If you answered yes, what port was it on?

→ yes, the switch displayed the MAC address of pc-A, which is ~~5c~~ 5c26.0a24.2a60. It was on port Fa0/6.

Reflection Questions:

1, Can you have broadcasts at the layer 2 level?
If so, what would the MAC address be?

→ yes, broadcasts can occur at layer 2 level.
The MAC address for a broadcast is 'ff:ff:ff:ff:ff:ff'. This address is used to send a packet to all devices on local network segment.

2, why would you need to know the MAC address of a device?

→ knowing the MAC address of a device is important for several reasons:

Device Identification: To uniquely identify devices on the network.

- Network management: To configure and manage network access controls, such as MAC filtering.

Troubleshooting: To diagnose network issues and ensure packets are reaching the correct devices.

- Security: To track and authenticate devices connecting to the network and to prevent unauthorized access.

DISCUSSION:

→ This lab exercise provided a comprehensive understanding of configuring network devices, verifying connectivity and analyzing MAC addresses on a switch and a PC. Key steps included the configuring IP addresses, verifying connectivity using

ping commands and examining MAC address tables. Initially, both the switch(S1) and PC-A were configured with IP addresses 192.168.1.2 and 192.168.1.3, respectively. The successful pings between them confirmed proper network configuration and the connectivity.

The Burned-In Address (BIA) was examined, highlighting its importance for device identification. The commands like 'show mac address-table' and 'show interfaces vlan 1' were essential for network management and troubleshooting, providing the practical experience in understanding MAC addresses, device identification and network security.

CONCLUSION :-

This lab successfully demonstrated the configuration of network devices, verification of the connectivity and analysis of MAC addresses. By configuring IP addresses & using commands to retrieve MAC address information, we gained the valuable insights into network communication at the data link layer. The successful ping tests confirmed that the network was correctly set up, and the analysis of MAC addresses provided a deeper understanding of device identification and network management. Overall, the lab provided practical experience in configuring and managing a network highlighting the importance of MAC addresses in ensuring effective and secure network communication.



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR



Lab Report No:- 4

Title:- Observe DNS Resolution

Submitted by:-

Name:- Swaraj Shrestha

Roll No:- 38

Submitted To:-

Department Of CSIT

Checked by:-

Shrestha

Date of submission:- 2081/04/18

TITLE: Observe DNS Resolution.

OBJECTIVE:-

- To observe the DNS conversion of a URL to an IP address.
- To use the nslookup command for DNS lookup on a website.
- To use the nslookup command for DNS lookup on mail servers.

Resource Required:

→ 1 PC (Windows with Internet and cmd).

THEORY:

→ The Domain Name System (DNS) is hierarchical system used to translate domain names, which are easy for humans to remember, into IP addresses, which are used by computers to identify each other on the network. DNS allows users to enter domain names into their web browsers rather than having to remember IP addresses.

When a URL is typed into a web browser, the DNS process is initiated to resolve the domain name into an IP address. For instance, typing "www.cisco.com" requires DNS to convert this domain into an IP address so that the browser can locate the Cisco web servers.

DNS uses a distributed database to store domain names and their corresponding IP addresses. Each part of the URL serves a specific purpose, with the protocol (such as HTTP or HTTPS) indicating how data should be transferred and the domain name indicating the specific resource to be accessed.

OBSERVATIONS:

1. part 1:

- Pinging "www.icann.org" resolved to both IPv6 and IPv4 addresses.
- Entering the addresses directly into a web browser successfully displayed the ICANN home page.

```
C:\> ping www.icann.org
```

```
Pinging www.vip.icann.org [2620:0:2d0:200::7] with 32 bytes of data:  
Reply from 2620:0:2d0:200::7: time=43ms  
Reply from 2620:0:2d0:200::7: time=41ms  
Reply from 2620:0:2d0:200::7: time=44ms  
Reply from 2620:0:2d0:200::7: time=39ms
```

```
Ping statistics for 2620:0:2d0:200::7:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 39ms, Maximum = 44ms, Average = 41ms
```

Q, when you ping www.cisco.com, do you get the same IP address as the example? Explain.

→ when you ping www.cisco.com, you might not get the same IP address as the example provided in lab. This occurs because large websites

like cisco often use a network of mirrored or load-balanced servers distributed across the different geographic locations. This distribution ensures better performance, redundancy and availability by directing users to the server closest to them or one that is currently under the least load.

factors :-

- a, Geographic Distribution: Companies like cisco utilize content Delivery Networks (CDNs) and multiple data centers around the world. The DNS system directs your request to nearest or most appropriate server based on various factors like your location & current server load.
- b, Load Balancing: DNS uses load balancing techniques to distribute incoming traffic across multiple servers. This helps in preventing any single server from becoming overwhelmed & ensures that users have a smooth and efficient browsing experience.
- c, Dynamic DNS updates: The IP addresses associated with Domain names can change over time due to DNS updates. This can be due to changes in the network infrastructure, server maintenance or optimization strategies implemented by the hosting provider.

Type the IP address that you obtained when you pinged `www.cisco.com` into a browser. Does the website display? explain.

→ The IP address obtained is `72.163.4.185`. When I entered the IP obtained from pinging `www.cisco.com` into a web browser, the website may not display correctly. This is because many large websites, including Cisco, use virtual hosting and SSL certificates tied to their domain names. Entering the IP address directly bypasses the domain name, which can result in the server not knowing which website to display or SSL/TLS errors.

> Part 2:

- The default DNS server was identified.
 - `nslookup` for "`www.cisco.com`" returned multiple IP addresses, indicating mirrored servers.
- Translating the IP address back into a domain name showed the domain's corresponding name.

```
> www.cisco.com
Default Server: 1.1.1.1
Address: 1.1.1.1

Non-authoritative answer:
Name:   e2867.dsca.akamaiedge.net
Addresses: 2600:1404:a:395::b33
           2600:1404:a:38e::b33
           172.230.155.162

Aliases: www.cisco.com
         www.cisco.com.akadns.net
         wwwds.cisco.com.edgekey.net
         wwwds.cisco.com.edgekey.net.globalredir.akadns.net
```

IK.

25

2, what is the translated IPv4 address?

→ The translated IPv4 address for the `www.cisco.com` is `172.230.155.162`.

2, Is it the same as the IP address shown with the `ping` command?

→ yes, the translated IPv4 address `172.230.155.162` is the same as the IP address shown with the `ping` command.

3, Part 3:

- nslookup revealed multiple mail servers for "cisco.com" with different MX preferences.
- The server with the lowest MX preference would be contacted first for email delivery.

```
> cisco.com
```

```
Server: one.one.one.one
```

```
Address: 1.1.1.1
```

```
Non-authoritative answer:
```

```
cisco.com      MX preference = 20, mail exchanger = rcdn-mx-01.cisco.com
```

```
cisco.com      MX preference = 30, mail exchanger = aer-mx-01.cisco.com
```

```
cisco.com      MX preference = 10, mail exchanger = alln-mx-01.cisco.com
```

Q, what is the fundamental purpose of DNS?

→ The fundamental purpose of DNS is to translate human-friendly domain names (like `www.example.com`) into machine friendly IP addresses (like `192.0.2.1`), allowing users to access websites and other resources on the Internet using easy to remember names instead of numerical IP addresses.

DISCUSSION:-

→ In this lab, three main activities were performed to understand DNS resolution:

part 1: Observe the DNS conversion of a URL to an IP Address.

ping the URL: using the command prompt, the URL for the Internet Corporation for Assigned Names and Numbers (ICANN) was pinged to observe the translation of "www.icann.org" into an IP address.

Browser Test: The translated IP address was entered into a web browser to verify that the ICANN home page could be accessed without using the domain name.

ping Additional URL: The process was repeated for "www.cisco.com" to see if the IP address resolved was consistent.

2: Observe DNS Lookup using the nslookup command on a website.

- Default DNS server: The nslookup command was used to find the default DNS server and perform DNS lookups for "www.cisco.com".
- Translate Domain to IP: The nslookup tool was used to translate domain names into IP address and vice versa.

part 3: Observe DNS Lookup using the nslookup command on mail servers.

mail server lookup: The nslookup command was used with the "set type = mx" parameter to identify mail servers for the domain "cisco.com".

Redundancy: The concept of redundancy in mail servers was observed, ~~noting~~ noting that the multiple mail servers with different MX preferences ensure reliable email delivery.

CONCLUSION:

→ This lab demonstrated the essential role of DNS in network communications by converting human-friendly domain names into machine-friendly IP addresses. By using commands like ping and nslookup, we can observe and understand the DNS resolution process, verify DNS functionality, and troubleshoot potential network issues. The redundancy in DNS and mail servers ensures robustness and reliability in network communications, highlighting the importance of DNS in maintaining seamless internet and email services.



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR

Lab Report No:- 5

Title:- Building a Basic Router & Switch Network.

Submitted by:-

Name:- Surajan Shrestha

Roll No:- 38

Submitted To:-

Department Of CSIT

Checked by:-

Shrestha

Date of submission:- 2081/04/25

TITLE: Building a Basic Router & Switch Network

OBJECTIVE:

→ To create a network topology using a Cisco 1941 Router and a switch packet Tracer, configure the network devices, and verify the network connectivity.

MATERIALS - REQUIRED :

- Cisco Packet Tracer Software,
- Cisco 1941 Router,
- Cisco 2960 Switch &
- Two PCs.

THEORY :-

→ In this lab, we aim to build a basic network by interconnecting a router, switch, and PCs. The router serves as the central device that connects different networks, while the switch is used to connect devices within the same network. By configuring IP addresses on the router's interfaces and setting up static routes, we establish the communication between devices across different subnets.

Key - Concepts :-

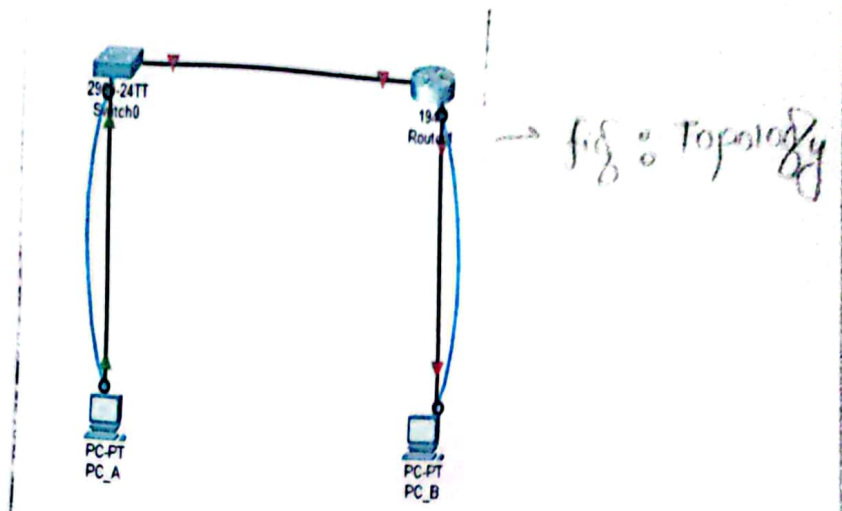
1. Router : A network device that forwards data packets between computer networks. It performs the traffic directing functions on the internet & is crucial in enabling communication between different network segments.

2. Switch :- A networking device that connects the devices within a single network. It operates at the data link layer (layer 2) of the OSI model & is responsible for forwarding data based on mac addresses.
3. FastEthernet Interfaces :- These are ports on the routers and switches used to connect devices using Ethernet cables. Unlike GigabitEthernet, which supports higher data rates, FastEthernet supports speeds up to 100 mbps.
4. IP Addressing :- IP addresses are unique identifiers for devices on a network. They are used to route data to the correct destination. In this lab, each device is assigned a static IP address.
5. Subnetting :- Subnetting involves dividing a network into smaller, more manageable sub-networks (subnets). Each subnet is identified by a subnet mask.

PROCEDURE :-

- Step-1 :- Set up the Topology in Cisco Packet Tracer
- a. Add Devices :- Add cisco 1941 router, one cisco 2960 switch and two pc's to the workspace.
- b. Connect devices :-
- Connect the router's 'fastEthernet0/0' to the switch's 'fastEthernet0/1' using a copper Straight-Through cable.

- Connect PC-A to switch's 'fastEthernet0/2' and PC-B to the router's 'fastEthernet0/1'.



step - 2 : Configure x Devices

1. Configure PCs:

- Assign IP addresses to PC-A & PC-B.
- PC-A: IP Address '192.168.1.3', Subnet Mask '255.255.255.0', Default Gateway '192.168.1.1'.



PC_A

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

- PC-B : Ip address '192.168.0.3', subnet mask '255.255.255.0', Default Gateway '192.168.0.1'



PC_B

Physical Config Desktop Programming Attributes

P Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.0.3

Subnet Mask 255.255.255.0

Default Gateway 192.168.0.1

DNS Server 0.0.0.0

2, Configure the Router

→ Console into the router and enable the privileged EXEC mode.

Router > enable

Router# config terminal

Router(config)# hostname R1

R1(config)# no ip domain lookup

R1(config)# enable secret class

R1(config)# line console 0

R1(config-line)# password cisco

R1(config-line)# login

R1(config)# line vty 0 4

R1(config-line)# password cisco

R1(config-line)# login

R1(config)# service password-encryption

R1(config)# banner motd \$ Authorized users only \$

- Configure & activate both interfaces on router.
- ```

→ R1(Config)# interface GigabitEthernet0/0.
R1(Config-if)# ip address 192.168.0.1 255.255.
255.0
R1(Config-if)# ipv6 address 2001:db8::dead::1/64
R1(Config-if)# ipv6 address FE80::%1 link-local
R1(Config-if)# no shutdown
R1(Config-if)# exit

```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```

R1>enable
R1>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(Config)#no ip domain-lookup
R1(Config)#enable secret class
R1(Config)#line console 0
R1(Config-line)#password class
R1(Config-line)#password class
R1(Config-line)#login
R1(Config-line)#line vty 0 4
R1(Config-line)#password class
R1(Config-line)#login
R1(Config-line)#config terminal
R1(Config)#
R1(Config)#service password-encryption
R1(Config)#banner motd 'Authorized User Only'
R1(Config)#interface g0/0/0
R1(Config-if)#
R1(Config-if)#interface GigabitEthernet0/0
R1(Config-if)#ip address 192.168.0.1 255.255.255.0
R1(Config-if)#ipv6 address 2001:db8::dead::1/64
R1(Config-if)#ipv6 address FE80::%1 link-local
R1(Config-if)#no shutdown
R1(Config-if)#
*LINE-5-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(Config)#interface GigabitEthernet0/1
R1(Config-if)#ip address 192.168.1.1 255.255.255.0
R1(Config-if)#ipv6 address 2001:db8::dead::1/64
R1(Config-if)#ipv6 address FE80::%1 link-local
R1(Config-if)#no shutdown
R1(Config-if)#
*LINE-5-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R1(Config-if)#
R1(Config)#

```

Copy Paste

Top

- Configure an interface description for each interface indicating which device is connected to it.
- ```

R1(Config)# interface GigabitEthernet 0/1
R1(Config-if)# description Connected to fastEthernet0/5
on s1
R1(Config-if)# exit
R1(Config)# interface GigabitEthernet0/0
R1(Config-if)# description Connected to Host PC-B
R1(Config-if)# exit

```


R1 (config)# ipv6 unicast-routing

R1 (config)# exit

R1 # copy running-config startup-config

R1 # clock set 15:30:00 27 Aug 2019

```
R1(config-if)#interface GigabitEthernet0/1
R1(config-if)#description Connected to FastEthernet0/5 on S1
R1(config-if)#exit
R1(config)#interface GigabitEthernet0/0
R1(config-if)#description Connected to Host PC_3
R1(config-if)#exit
R1(config)#ipv6 unicast-routing
R1(config)#exit
R1#
*SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#clock set 15:30:00 27 Aug 2019
R1#
```

Copy

Paste

3, Configure the switch

↳ commands :

Switch > enable

Switch# config terminal

Switch(config)# hostname S1

~~switch~~ S1(config)# no ip domain-lookup

S1(config)# interface vlan 1

~~S1 (config)#~~

S1 (config-if)# ip address 192.168.1.2 255.
255.255.0

S1 (config-if)# no shutdown

S1 (config-if)# exit

S1 (config)# ip default-gateway 192.168.1.1

S1 (config-if)# exit.

Physical Config (2) Attributes

```

# Cisco IOS Software, C1900 Software (C1900-LANBASE-M), Version 17.1(2)DN, RELEASE SOFTWARE (P)
Copyright (c) 1986-2018 by Cisco Systems, Inc.
Compiled Wed 12-Oct-18 22:08 by po_team

Press RETURN to get started!

%LINK-3-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINK-3-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up

Switch#enable
Switch#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#no ip domain-lookup
S1(config)#interface vlan 1
S1(config-if)#ip address 192.168.1.2 255.255.255.0
S1(config-if)#no shutdown

% Invalid input detected at '' marker.

S1(config-if)#no shutdown

S1(config-if)#
%LINK-3-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
S1(config)#ip default-gateway 192.168.1.1
S1(config)#exit
S1#
SYS-S-CFGIS_1: Configured from console by console

```

step:- step - 3:- Display Device Information

- 1, Displaying routing table on router.
↳ R1# show ip route

```

R1#enable
Password:
Password:
Password:
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, IA - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.0.0/24 is directly connected, GigabitEthernet0/0
C    192.168.0.1/32 is directly connected, GigabitEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/1
C    192.168.1.1/32 is directly connected, GigabitEthernet0/1

R1#

```

Paste

Qsn:

a, what code is used in routing table to indicate a directly connected network?

→ The code used in routing table to indicate a directly connected network is 'c'.

b, How many route entries are coded with a 'c' code in routing table?

→ If we assume that the router has 2 directly connected interfaces (as configured in the lab report):

- 'GigabitEthernet 0/0/0' with ip '192.168.0.1/24'

- 'GigabitEthernet 0/0/1' with ip '192.168.1.1/24'

the router/routing table would typically have 2 route entries coded with a 'c' code.

c, what interface types are associated with the 'c' coded routes?

→ The interface types associated with the 'c' coded routes would be:

- 'GigabitEthernet 0/0/0'

- 'GigabitEthernet 0/0/1'

These interfaces correspond to the two networks that are directly connected to the Cisco 1941 router.

2, Using the show ipv6 route command on Router R1 to display the IPv6 routes.

→ Command:

R1# show ipv6 route

```

R1#show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
C 2001:DB8:ACAD::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L 2001:DB8:ACAD::1/128 [0/0]
    via GigabitEthernet0/0, receive
C 2001:DB8:ACAD::1::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L 2001:DB8:ACAD::1::1/128 [0/0]
    via GigabitEthernet0/1, receive
L FF00::/8 [0/0]
    via Null0, receive
R1#

```

Copy

Paste

2, Displaying Interface Information on Router R1.
a, use the show ip interface `g0/0/1` & :

```

R1#show ip interface GigabitEthernet0/1
GigabitEthernet0/1 is up, line protocol is up (connected)
Internet address is 192.168.1.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachable are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/RTCP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: MCI Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled

```

Q3n:

a, what is the operational status of the `G0/0/1` interface?

→ The operational status of the `GigabitEthernet0/0/1` interface can be checked using the 'show ip interface brief' command. The status is typically

shown as :

- status : 'up' (indicating that the interface is enabled and operational)
- protocol : 'up' (indicating that the line protocol is operational)

So, the operational status is 'up' and 'up'.

→ what is the MAC address of the G0/1 interface?

→ The MAC address of the 'GigabitEthernet0/0/1' interface can be found using 'show interface gigabit-ethernet0/0/1' command and the mac address is 0019.e86a.1234 .

→ What is the Internet address displayed in 'show ip interface brief' command?

→ The Internet address is displayed in the 'show interfaces gigabitEthernet0/0/1' command output under a line similar to:

- Internet address is '192.168.1.1/24'.

This line shows the assigned ip address and subnet mask in CIDR notation.

Example command output:

```
R1#
R1#
R1#
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.0.1    YES manual up          up
GigabitEthernet0/1  192.168.1.1    YES manual up          up
Vlan1          unassigned     YES unset  administratively down down
R1#show ipv6 interface brief
GigabitEthernet0/0  [up/up]
FE80::1
2001:DB8:ACAD::1
GigabitEthernet0/1  [up/up]
FE80::1
2001:DB8:ACAD::1::1
Vlan1              [administratively down/down]
unassigned
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.0.1    YES manual up          up
GigabitEthernet0/1  192.168.1.1    YES manual up          up
Vlan1          unassigned     YES unset  administratively down down
R1#
```

an/ys22

Reflection Questions:

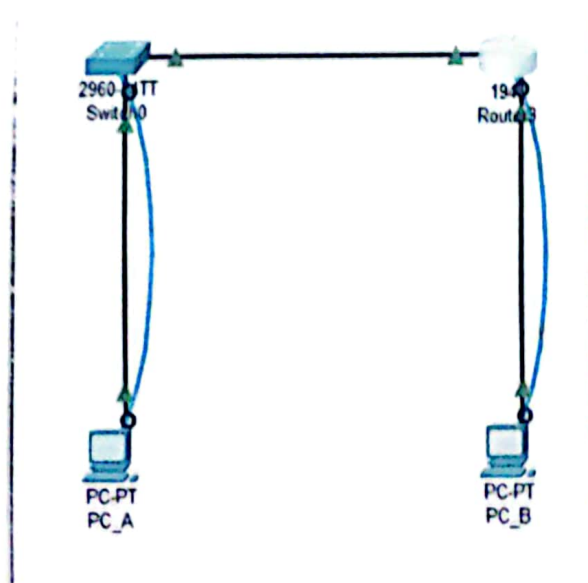
1. If the G0/0/1 interface showed that it was administratively down, what interface configuration command would you use to turn the interface up?

→ I would use command `RL (config-if) # no shutdown`.

What would happen if you had incorrectly configured interface G0/0/1 on the router with an IP address of 192.168.1.2?

→ PC-A would not be able to ping PC-B. This is because PC-B is on a different network than PC-A which requires the default gateway router to route these packets. PC-A is configured to use the IP address of 192.168.1.1 for the default-gateway router, but this address is not assigned to any device on the LAN. Any packets that need to be sent to default-gateway for routing will never reach their destination.

final-topology :-



DISCUSSION:

The lab demonstrates the fundamental concepts of network topology design, router configuration and IP addressing. By configuring the router with appropriate IP addresses & enabling the interfaces, the router successfully routes traffic between two different subnets. The switch facilitates communication within the same network, ensuring that devices connected to it can reach each other.

The challenge faced during the lab was the initial attempt to configure non-existent the Gigabit Ethernet interfaces on the Cisco 1941 router, leading to an "Invalid interface type & number" error. This was resolved by correctly using Fast Ethernet interfaces and do on.

CONCLUSION:

→ This lab was successfully completed with a functional network topology established between a Cisco 1941 router, a switch and two PCs. The PCs were able to communicate across different subnets, verifying the router was correctly configured to route traffic between the networks. This lab reinforces the understanding of basic router and switch configurations, IP addressing, and network connectivity testing.



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY



HIMALAYA COLLEGE OF ENGINEERING
CHYASAL, LALITPUR

Lab Report No:- 6

Title:- Introduction to network protocol
Analysis

Submitted by:-

Name:- Surajan Shrestha

Roll No:- 38

Submitted To:- Ram Krishna Paul

Department Of CSIT

Checked by:-

Ram Krishna

Date of submission:- 2081/05/08

TITLE: Introduction to Network Protocol Analysis.

OBJECTIVE:

→ To observe network protocols in action by capturing and analyzing packets using the Wireshark packet analyzer. This lab will introduce the basic functions of Wireshark and demonstrate how to capture and analyze network traffic, such as HTTP requests.

THEORY:

→ Wireshark is a widely used packet analyzer that allows users to capture and inspect the details of messages exchanged over the network. It operates by receiving copies of packets transmitted or received by the computer's network interfaces and displays their detailed structure. Wireshark can analyze various network protocols and works on Windows, Mac, and Linux/Unix platforms. The program consists of two key components:

1, Packet Capture Library: This module receives all the link-layer frames (packets) sent from or received by a computer over a specific network interface (e.g., ethernet, wifi). These captured packets contain all the messages exchanged by various protocols such as HTTP, FTP, TCP, UDP and IP.

2. Packet Analyzer: It analyzes and displays the contents of the captured packets. The analyzer can parse Ethernet frames to extract IP datagrams, TCP segments and the application-layer messages like HTTP, thus enabling detailed packet inspection.

Wireshark's features:

- Wireshark captures frames across different network layers.
- It can filter packet data based on the protocol or other characteristics.
- The tool provides an intuitive GUI with five major components:
 1. Command menus
 2. Packet-listing window
 3. Packet-Header Details window
 4. Packet-Contents window
 5. Packet Display filter field.

Installation of Wireshark:

→ To run Wireshark, you must first install the software and the appropriate packet capture library (libpcap or WinPcap) follow these steps:

- 1, visit the Wireshark download page and download the appropriate version for your operating system.
- 2, Install the Wireshark binary, which will also install libpcap or WinPcap.

wireshark In Action : Once installed, wireshark can be run to capture network traffic and inspect it. The captured packets can reveal the operations of various protocols at different layers, from the link layer (Ethernet/WiFi) to application protocols like HTTP.

In this lab, you will :

- capture packets using wireshark.
- Use a web browser to generate HTTP requests.
- Analyze the captured traffic by applying the filters.

Procedure :

1, Start wireshark :

- open wireshark and choose the active network interface (Ethernet or WiFi) for capturing traffic.

2, Capture Network Traffic :

- start capturing packets by clicking the interface and let wireshark run.

3, visit websites :

- open your browser and visit bbc.com and ican.com to generate network traffic involving multiple protocols such as DNS, HTTP/HTTPS, TCP and TLS.

4, Stop packet capture :

- once the pages have loaded, stop the capture by clicking the red square button.

5, Apply filters :

- Use the following filters in the display filter box to isolate different types of the network traffic :
 - ICMP : Type icmp to see ping or network diagnostic packets.
 - TCP : Type tcp to isolate TCP packets.
 - DNS : Type dns to show DNS requests and responses.
 - HTTP : Type http to filter HTTP packets.
 - TLS : Type tls to view encrypted traffic using HTTPS.
 - UDP : Type udp to capture UDP packets.
 - Broadcast Traffic : Type eth.dst == ff:ff:ff:ff:ff:ff to capture the broadcast traffic in the network.

6, Analyze packets :

- For each protocol filter, observe how the packets are structured, the source and destination IP addresses, the data exchanged, and any relevant protocol-specific details.
- Analyze broadcast traffic to observe how it is used for communication like ARP requests.

OBSERVATION :

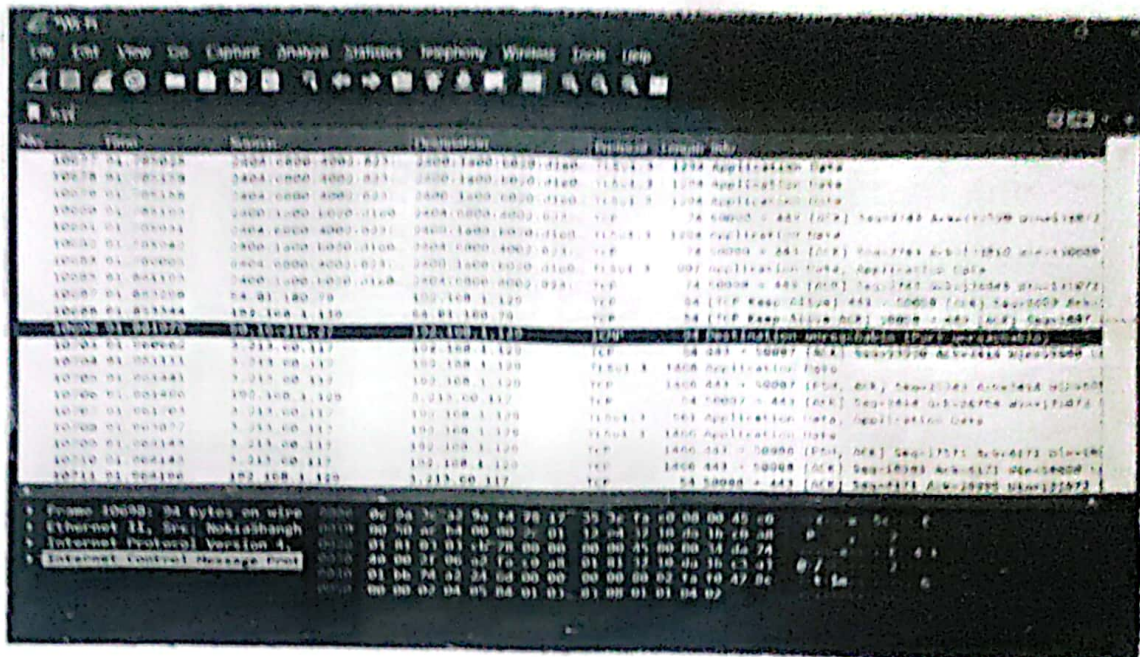
- Capturing packets from wifi :

No.	Time	Source	Destination	Protocol	Length	Info
358	20.994908	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.119? Tell 192.168.1.64
359	20.994908	57.128.101.79	192.168.1.129	TCP	54	[TCP Keep-Alive] 443 → 49578 [ACK] Seq=0 Ret=7 Window=0
360	20.994908	57.128.101.79	192.168.1.129	TCP	66	[TCP Keep-Alive ACK] 443 → 49578 [ACK] Seq=0 Ret=7 Window=0
361	20.994974	192.168.1.129	57.128.101.79	TCP	54	[TCP Dup ACK 243] 40678 → 443 [ACK] Seq=2 Ret=1 Window=0
362	21.300333	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	ICMPv6	86	Neighbor Solicitation for 2400:1a00:b020:d1e0:1 from 2400:1a00:b020:d1e0:2
363	21.300333	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	ICMPv6	86	Neighbor Solicitation for 2400:1a00:b020:d1e0:1 from 2400:1a00:b020:d1e0:2
364	21.300398	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	ICMPv6	86	Neighbor Advertisement 2400:1a00:b020:d1e0:2 (no) 1
365	21.300457	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	ICMPv6	86	Neighbor Advertisement 2400:1a00:b020:d1e0:2 (no) 1
366	21.302811	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	ICMPv6	86	Neighbor Solicitation for 2400:1a00:b020:d1e0:1 from 2400:1a00:b020:d1e0:2
367	21.302870	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	ICMPv6	86	Neighbor Advertisement 2400:1a00:b020:d1e0:2 (no) 1
368	21.535480	2400:1a00:b020:d1e0...	2606:4700:4400::681	TCP	75	49957 → 443 [ACK] Seq=1 Ack=1 Win=512 Len=1 TTFB exp=0
369	21.540229	2606:4700:4400::681	2400:1a00:b020:d1e0...	TCP	86	443 → 49957 [ACK] Seq=1 Ack=2 Win=12 Len=0 SLS=1 SR=0
370	21.727435	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 169.254.169.254? Tell 192.168.1.129
371	22.016383	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.119? Tell 192.168.1.64
372	22.221706	NokiaShangha_3c:fa:...	Intel_a2:9a:f4	ARP	60	Who has 192.168.1.129? Tell 192.168.1.64
373	22.221731	Intel_a2:9a:f4	NokiaShangha_3c:fa:...	ARP	42	192.168.1.129 is at 0c:9a:3c:fa:2:9a:f4
374	22.763540	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 169.254.169.254? Tell 192.168.1.129
375	23.734345	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 169.254.169.254? Tell 192.168.1.129
376	24.723764	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 169.254.169.254? Tell 192.168.1.129

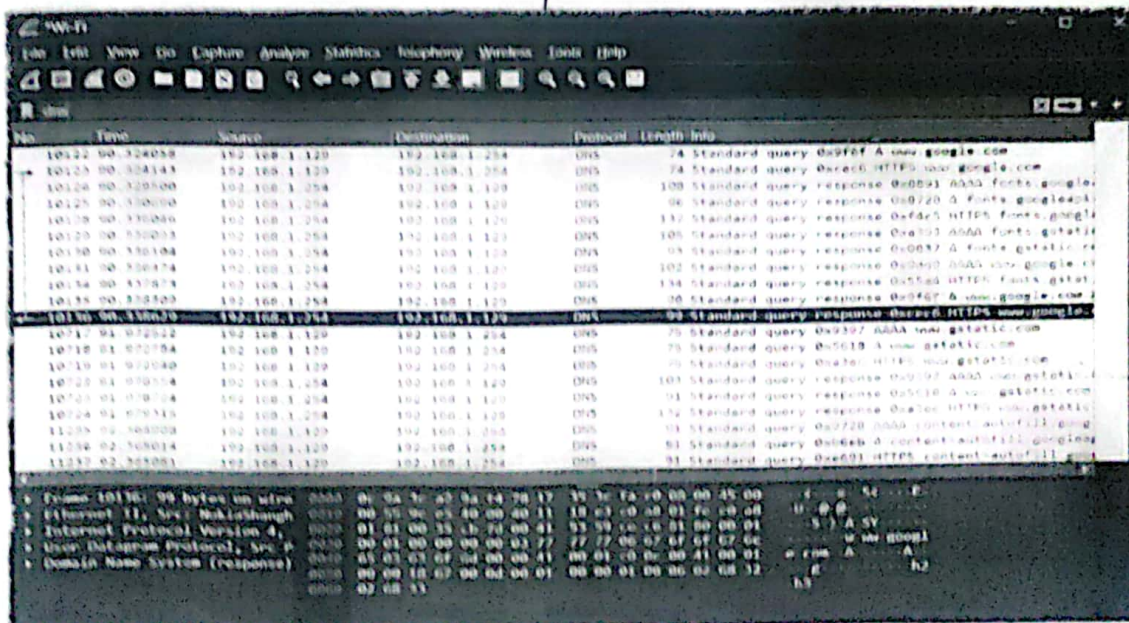
- ICMP Traffic : observing ICMP packets the reveals pings on error messages like the "Destination unreachable" and timestamps.

No.	Time	Source	Destination	Protocol	Length	Info
98	...	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9880	80.532380	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9883	80.762565	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9891	81.025978	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9895	81.261345	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9903	81.509259	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9906	81.769908	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9908	82.035654	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9915	82.639414	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
9933	83.776946	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
10010	88.604503	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
10041	88.871356	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
10062	89.614258	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
10078	89.909815	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
10447	91.621430	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)
10698	91.881979	50.16.218.27	192.168.1.129	ICMP	94	Destination unreachable (Port unreachable)

- TCP packets : Tcp traffic includes handshakes (SYN, ACK) and data exchange.



- DNS Queries: You will observe requests to resolve domain names (e.g. `bbc.com`) into IP addresses and responses from DNS servers.



- HTTP/TLS (HTTPS) : HTTP packets capture web communication in plain text, while TLS capture encrypted web traffic.
- for TLS :

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

udp

No.	Time	Source	Destination	Protocol	Length	Info
9873	80.064258	2400:1a00:b020:d1e0...	2404:6800:4002:82c...	QUIC	94	Protected Payload (KFO), DCID=fca2b08123f9b077
9875	80.123115	192.168.1.254	192.168.1.129	DNS	80	Standard query response 0x3ff2 A www.ican.com
9877	80.488792	192.168.1.254	192.168.1.129	DNS	126	Standard query response 0xb10 HTTPS www.ican.com
9878	80.488792	192.168.1.254	192.168.1.129	DNS	126	Standard query response 0xb40f AAAA www.ican.com
9884	80.979609	192.168.1.129	192.168.1.254	DNS	72	Standard query 0xb130 AAAA www.ican.com
9885	80.979785	192.168.1.129	192.168.1.254	DNS	72	Standard query 0xf1ab A www.ican.com
9886	80.979850	192.168.1.129	192.168.1.254	DNS	72	Standard query 0x49c7 HTTPS www.ican.com
9887	80.989139	192.168.1.254	192.168.1.129	DNS	88	Standard query response 0xf1ab A www.ican.com
9888	80.989279	192.168.1.254	192.168.1.129	DNS	126	Standard query response 0xb130 AAAA www.ican.com
9889	80.989533	192.168.1.254	192.168.1.129	DNS	126	Standard query response 0x49c7 HTTPS www.ican.com
9995	88.563784	2400:1a00:b020:d1e0...	2404:6800:4002:82c...	QUIC	213	Protected Payload (KFO), DCID=fca2b08123f9b077
9996	88.566316	2400:1a00:b020:d1e0...	2404:6800:4002:82c...	QUIC	1143	Protected Payload (KFO), DCID=fca2b08123f9b077
9997	88.566151	2400:1a00:b020:d1e0...	2404:6800:4002:82c...	QUIC	214	Protected Payload (KFO), DCID=fca2b08123f9b077
9998	88.570399	2400:1a00:b020:d1e0...	2404:6800:4002:82c...	QUIC	519	Protected Payload (KFO), DCID=fca2b08123f9b077
9999	88.570930	192.168.1.129	192.168.1.254	DNS	77	Standard query 0x993a AAAA www.digimedia.com
10000	88.571043	192.168.1.129	192.168.1.254	DNS	77	Standard query 0x036a A www.digimedia.com
10001	88.571134	192.168.1.129	192.168.1.254	DNS	77	Standard query 0x72b9 HTTPS www.digimedia.com
10002	88.593731	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	QUIC	90	Protected Payload (KFO)
10003	88.593780	2400:1a00:b020:d1e0...	2400:1a00:b020:d1e0...	QUIC	90	Protected Payload (KFO)

Frame 9889: 126 bytes on wire (1008 bytes captured) on interface 0
 Ethernet II, Src: NokiaShangh...
 Internet Protocol Version 4, Src: 192.168.1.254, Destination: 192.168.1.129
 User Datagram Protocol, Src Port: 53, Destination Port: 443
 Domain Name System (response)

0000	0c 9a 3c a2 9a f4 78 17	35 3c fa c0 08 00 45 00	...
0010	00 70 9a 80 40 00 40 11	1b 2d c0 a8 01 fe c0 a8	...
0020	01 81 00 35 f0 51 00 5c	a2 32 49 c7 81 80 00 01	...
0030	00 00 00 01 00 00 03 77	77 77 04 69 63 61 6e 03	...
0040	63 6f 6d 00 00 41 00 01	c0 10 00 06 00 01 00 00	...
0050	02 55 00 2a 03 6e 73 31	09 64 69 67 69 6d 65 64	...
0060	69 61 c0 15 03 64 6e 73	c0 2e 78 a5 2b 25 00 00	...
0070	2a 30 00 00 0e 10 00 09	3a 80 00 00 0e 10	...

Broadcast Traffic: Broadcast packets show network-wide messages, such as ARP requests used to resolve the MAC addresses.

Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ethernet II

No.	Time	Source	Destination	Protocol	Length	Info
3914	55.819851	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 192.168.1.129? Tell 192.168.1.129
3976	56.723784	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 192.168.1.129? Tell 192.168.1.129
4158	57.720882	Intel_a2:9a:f4	Broadcast	ARP	42	Who has 192.168.1.129? Tell 192.168.1.129
6491	60.007683	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
8148	61.032971	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
8464	62.056393	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
9055	70.650537	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
9333	71.560606	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
9363	72.604357	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
9381	80.002722	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
9984	81.710729	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
9918	82.742524	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
11016	116.227805	5a:02:fe:7e:1b:ac	Broadcast	ARP	42	Who has 192.168.1.254? Tell 192.168.1.129
13258	120.014883	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
20651	121.040908	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
20903	122.003161	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
21045	130.357721	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
21046	131.281731	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64
21069	132.407561	TpLinkTechno_18:44:...	Broadcast	ARP	60	Who has 192.168.1.110? Tell 192.168.1.64

Frame 9881: 60 bytes on wire (480 bytes captured) on interface 0
 Ethernet II, Src: TpLinkTechno...
 ARP Resolution Protocol (Request)

0000	ff ff ff ff ff ff 50 c7	b7 18 44 4d 00 00 00 01	...
0010	00 00 00 00 00 00 c0 c7	b7 18 44 4d c0 a8 01 40	...
0020	00 00 00 00 00 00 c0 c7	01 60 00 00 00 00 00 00	...
0030	00 00 00 00 00 00 00 00	00 00 00 00	...

DISCUSSION:

This lab demonstrates the utility of Wireshark in dissecting various protocols. Applying the filters narrows down the captured data, allowing deeper insights into protocol-

specific behavior. DNS queries are essential in browsing, as they resolve domain names into IP addresses, while TCP ensures reliable transmission. Broadcast packets are necessary for network-wide announcements, such as ARP and DHCP.

CONCLUSION :-

→ Wireshark provides powerful filtering options for inspecting network traffic. This lab helped capture and analyze specific protocols involved in web browsing activities. Understanding ICMP, TCP, DNS, HTTP, TLS, UDP, and broadcast traffic is critical in network troubleshooting and analysis.